MAULANA ABUL KALAM AZAD UNIVERSITY OF TECHNOLOGY

B.TECH. COMPUTER SCIENCE & ENGINEERING / ALLIED BRANCHES

END SEMESTER EXAMINATION (2024-25) -- SEMESTER 8

COURSE: DEEP LEARNING & NEURAL NETWORKS (PCC-CS801)

Time Allowed: 3 Hours Maximum Marks: 70 Marks Paper Code: PCC-CS801

Roll Number: [ ][ ][ ][ ][ ][ ][ ][ ][ ][ ][ ][ ][ ] | Total Printed Pages: 2

NOTE: Attempt all Sections. Assume suitable missing data if any. Programmable calculators are NOT permitted.

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SECTION A: Short Answer Type Questions (Attempt ALL questions) [10 x 2 = 20 Marks]

Q1. (a) Why do deep neural networks perform better than shallow networks on high-dimensional data?

Q1. (b) Explain vanishing and exploding gradient problems in deep architectures.

Q1. (c) What is Dropout regularization? How does it prevent overfitting during training?

Q1. (d) Explain convolution operation, stride, and padding (Valid vs Same) in CNNs.

Q1. (e) What is pooling layer in CNNs? Differentiate Max-Pooling and Average-Pooling.

Q1. (f) Explain difference between standard RNN, LSTM (Long Short-Term Memory), and GRU.

Q1. (g) What is Self-Attention mechanism in Transformer architecture?

Q1. (h) Define Generator and Discriminator loss functions in Generative Adversarial Networks (GANs).

Q1. (i) What is transfer learning? Explain fine-tuning vs feature extraction.

Q1. (j) Explain Adam (Adaptive Moment Estimation) optimizer equations.

SECTION B: Analytical & Computational Problems (Attempt any THREE) [3 x 10 = 30 Marks]

Q2. (a) Derive gradient update rule for weights in a convolutional layer using backpropagation.

Q2. (b) Explain LSTM internal architecture: Forget Gate, Input Gate, Candidate Memory, Output Gate with equations.

Q2. (c) Explain Transformer architecture encoder-decoder blocks and Multi-Head Scaled Dot-Product Attention.

Q2. (d) Explain Deep Residual Networks (ResNet) and why skip/residual connections allow training 100+ layer networks.

Q2. (e) Explain Generative Adversarial Networks (GAN) training loop, min-max game objective, and Nash equilibrium.

SECTION C: Long Technical & Design Questions (Attempt any ONE from each) [5 x 10 = 50 Marks]

Q3. (a) Explain CNN landmark architectures: LeNet-5, AlexNet, VGG-16, and Inception GoogLeNet. [10 Marks]

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(b) Explain Object Detection architectures: R-CNN, Fast R-CNN, Faster R-CNN, and YOLO (You Only Look Once). [10 Marks]

Q4. (a) Explain Sequence-to-Sequence (Seq2Seq) model with Bahdanau additive and Luong multiplicative attention. [10 Marks

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(b) Explain Bidirectional RNNs and Gated Recurrent Units (GRU) with reset and update gates. [10 Marks]

Q5. (a) Explain BERT (Bidirectional Encoder Representations from Transformers) pretraining: Masked LM and Next Sentence P

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(b) Explain GPT (Generative Pretrained Transformer) autoregressive causal language modeling and Reinforcement Learning from

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Q6. (a) Explain Variational Autoencoders (VAEs): Reparameterization trick, latent distribution, and ELBO objective. [10 M

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(b) Explain Diffusion Models (DDPM): Forward noise process, reverse denoising process, and score matching. [10 Marks]

Q7. (a) Explain batch normalization, layer normalization, group normalization, and weight decay. [10 Marks]

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(b) Explain deployment optimizations for deep models: Model quantization (INT8), pruning, and ONNX runtime. [10 Marks]

EXAMINATION EVALUATION SCHEME & RUBRIC GUIDELINES:

1. Section A: Direct definition, time complexity analysis, formula derivation, or syntax verification.
2. Section B: Step-by-step mathematical proof, trace tables, algorithmic pseudocode, and state diagrams.
3. Section C: Architectural design, full code implementation, edge case handling, and space/time tradeoffs.

***** END OF QUESTION PAPER -- DEVENGERS ACADEMIC VAULT *****